

Experiment - 2

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Design a UI where users recall visual elements (animal names hidden in a crossword). Evaluate the effect of chunking and short-term memory on user recall accuracy.

FRAME 1:

INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that divides information into smaller, meaningful units. The instruction screen of the Crossword Memory Game applies chunking effectively in the following ways:

1. Clear and Sequential Instructions

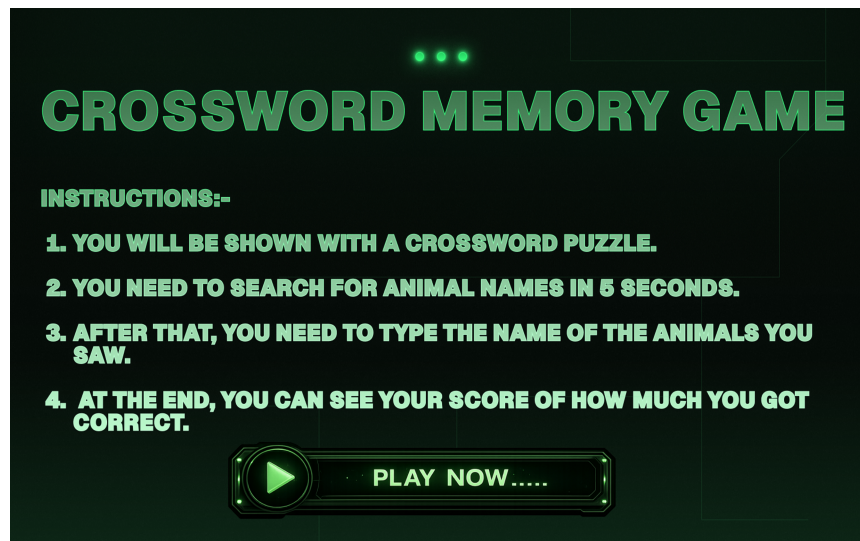
- Instructions are presented as numbered steps.
- Each step explains a single action, reducing cognitive overload.

2. Logical Information Flow

- Observation Phase – Users are informed about searching animal names.
- Time Constraint – A fixed time of 5 seconds is highlighted.
- Recall Phase – Users are told to type remembered animals.

3. Visual Hierarchy and UI Design

- Bold title immediately communicates the game purpose.
- Neon green text on dark background improves readability and focus.
- A glowing 'Play Now' button clearly indicates the next action.



FRAME 2:

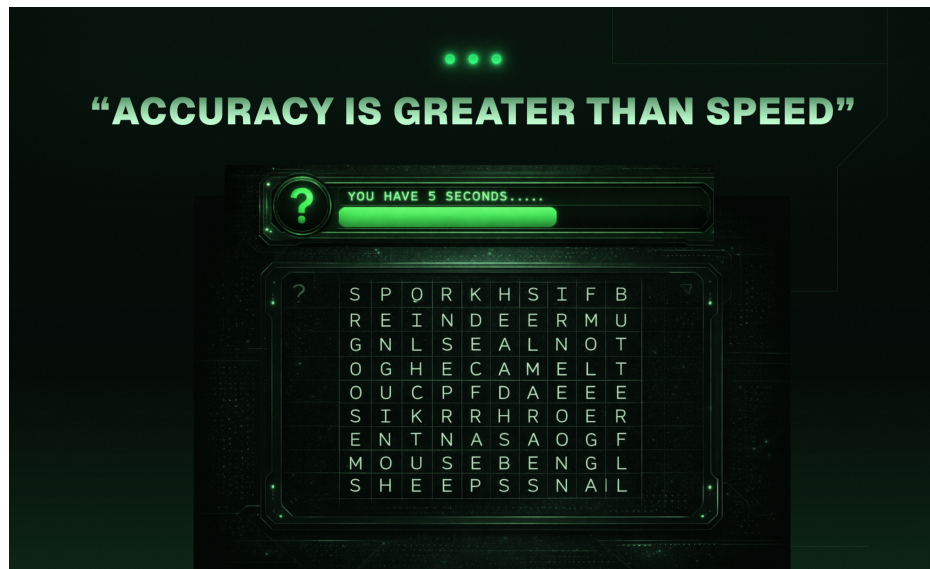
PUZZLE DISPLAY / CHUNKING PHASE:

This screen represents the chunking phase where users observe the crossword grid and mentally group animal names before recalling them. Key UI Elements and Analysis:

- Countdown Timer – Indicates limited viewing time and creates urgency.
- Crossword Grid – Letters are arranged in a structured grid to support visual scanning.
- Motivational Quote – Encourages accuracy over speed, promoting focused observation.

Chunking Benefits:

- Users group letters into recognizable animal names.
- Improves short-term memory encoding.
- Reduces visual clutter through grid alignment.



FRAME 3:

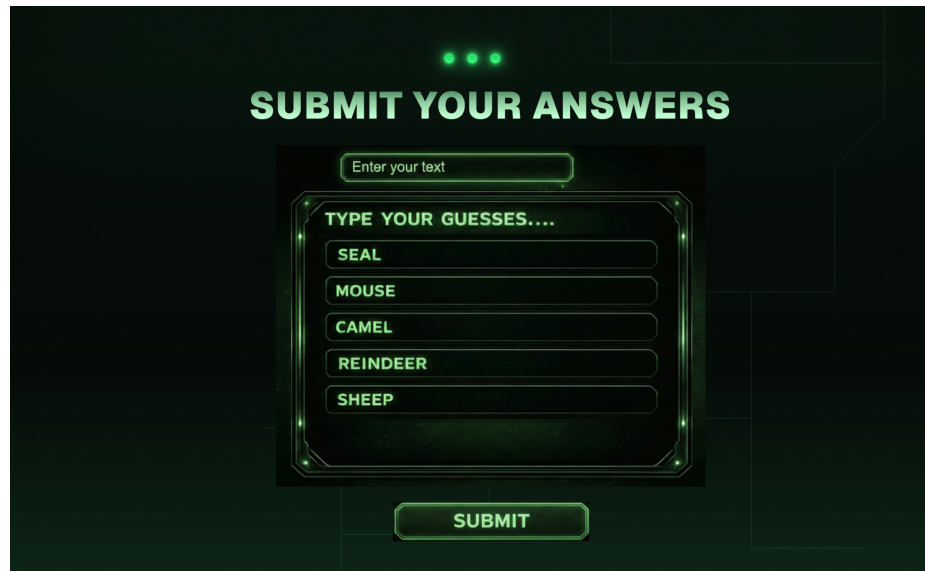
RECALL / ANSWER SUBMISSION PHASE:

This phase tests the user's ability to recall previously observed animal names without seeing the crossword again. UI Components and Functionality:

- Text Input Fields – Allow users to enter remembered animal names.
- Structured Layout – Maintains clarity and avoids confusion.
- Submit Button – Confirms the recall response.

Cognitive Impact:

- Evaluates short-term memory retention.
- Encourages focused recall rather than guessing.
- Reinforces the effectiveness of chunking.



FRAME 4:

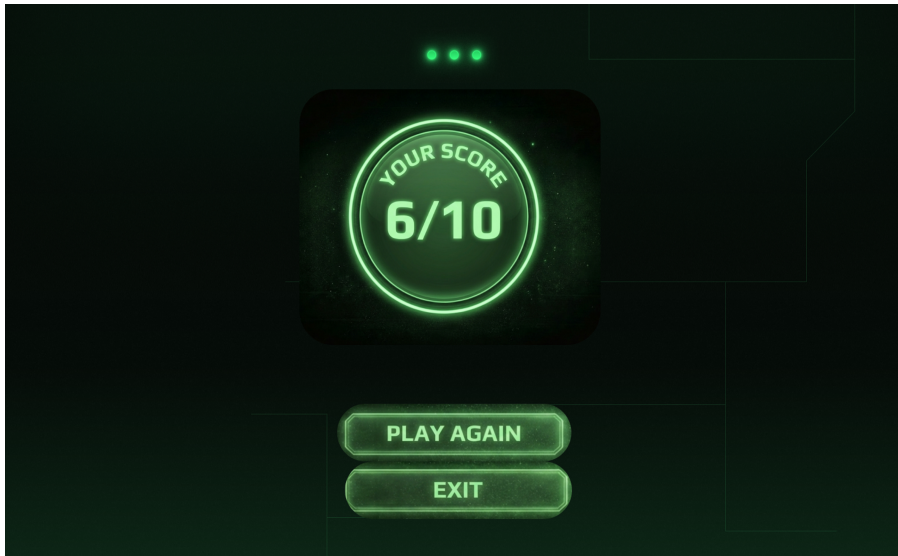
RESULT PAGE:

This screen displays the final score and provides feedback based on the user's recall accuracy. Key Features:

- Circular Score Indicator – Visually represents performance (e.g., 6/10).
- Play Again Option – Allows users to retry and improve memory.
- Exit Option – Ends the game session.

UX and Learning Benefits:

- Immediate feedback improves learning awareness.
- Gamified result presentation motivates replay.
- Clean and minimal design avoids stress.



Prototype link:-